

Diseño de un controlador para segundo orden

Ecuaciones principales

$$V_c(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} = R i_2(t) + R i_c(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s = R i_2 + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_c(s) = R I_1(s) + Ls [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$Ls [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2 + R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

$$V_c(s) = R I_1(s) + Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s)$$

$$V_c = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$V_s(s) = (R + \frac{1}{Cs}) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s) =$$

$$= \frac{Cs(Ls + 1 + 3RCs)}{Cs(Ls + R)} I_2(s)$$

$$V_s(s) = (R + \frac{1}{Cs}) I_2(s)$$

$$V_s(s) = \frac{RCs + 1}{Cs} I_2(s)$$

$$V_c(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_c(s) = (Ls + 2R) \frac{CLs^2 + 3RC + 1}{CLs^2 + RCs} I_2(s) - (Ls + R) I_2(s)$$

$$V_c(s) = \left[(Ls + 2R) \frac{CLs^2 + 3RC + 1}{CLs^2 + RCs} - (Ls + R) \right] I_2(s)$$

$$V_c(s) = \frac{CL^2s^2 + 3CLRs^2 + Ls + 2CLR^2s + 6CR^2s + 2R - CL^2s^2}{CLs^2 + RCs} I_2(s)$$

$$= \frac{CLs^2 + 3CLR^2s + 2R}{CLs^2 + RCs} I_2(s)$$

$$V_o(s) = \frac{3CLRs^2 + (9CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)$$

Funcion de transferencia

$$\frac{V_o(s)}{V_i(s)} = \frac{CRs + 1}{CS}$$

$$\frac{3CLRs^2 + (9CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{C^2LRs^2 + (CLs^2 + C^2R^2s^2 + CRs)}{3C^2LRs^2 + C^2(9R^2 + L)s + 2RC}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{Cs (CLRs^2 + Ls + CR^2s + R)}{Cs (3CRs^2 + 9C(9R^2 + L)s + 2R)}$$

$$\left[\frac{V_o(s)}{V_i(s)} = \frac{CLR s^2 + CL + CR^2 s + R}{3CLR s^2 + (9CR^2 + L)s + 2R} \right]$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s V_e(s) \left(1 - \frac{V(s)}{V_e(s)} \right)$$

V_e : escalon unitario (1V)

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR^2 + L) + R}{3CLRs^2 + (3CR^2 + L)s + 2R} \right) = 1 - \frac{R}{2R}$$

$$e(s) = 1 - \frac{1}{2} = \frac{1}{2} V$$

Estabilidad en lazo abierto

$$R = 1 \text{ k}\Omega$$

$$L = 47 \text{ }\mu\text{H}$$

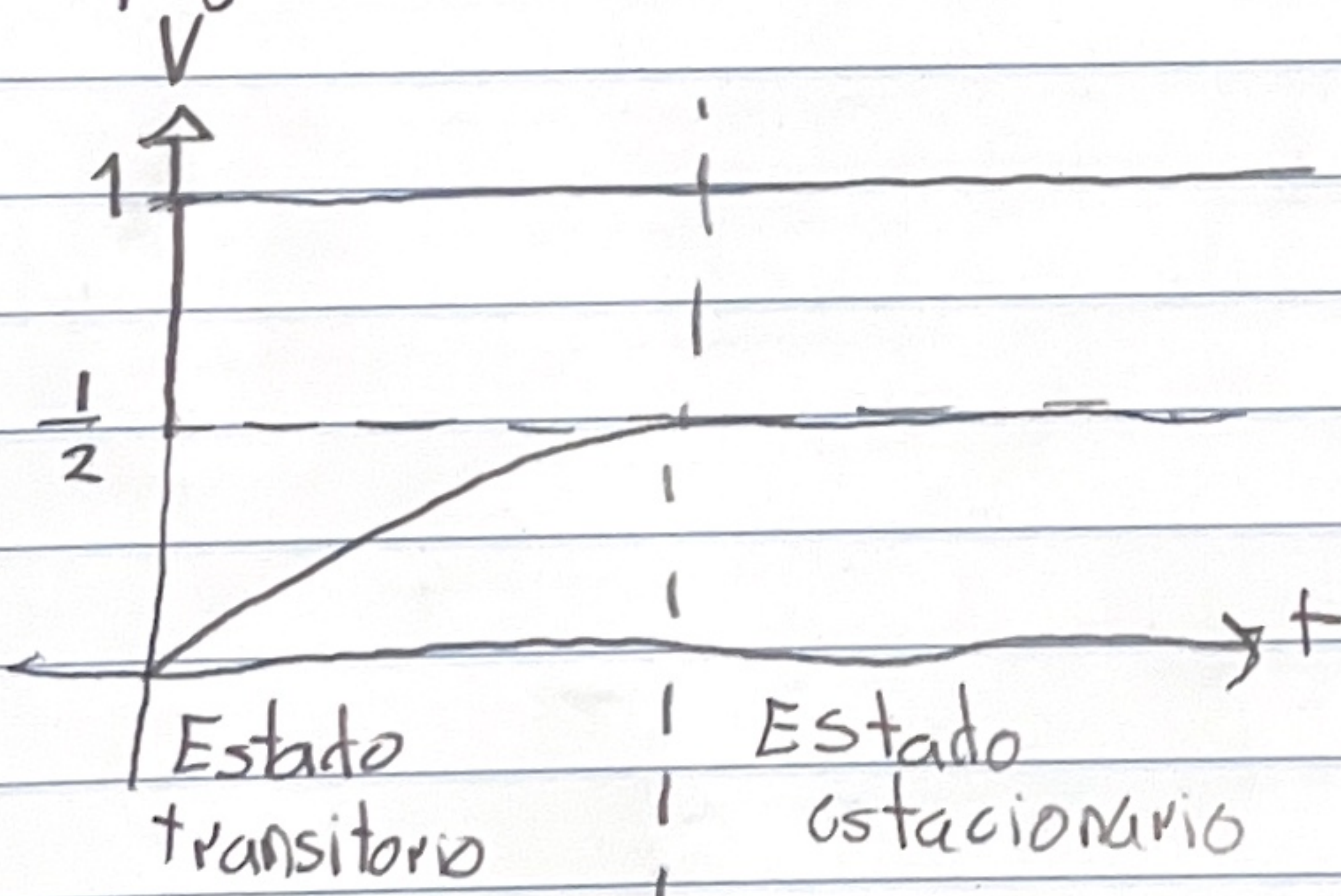
$$C = 470 \text{ }\mu\text{F}$$

$$\lambda_1 = -0.891063$$

$$\lambda_2 = -35460992.77$$

El sistema es estable con respuesta sobreamortiguada.

Bosquejo



Modelo de ecuaciones integrodiferenciales

El modelo se expresa despejando las variables dependientes, es decir, las corrientes

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{2} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{2} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_0(t) = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t))$$

$$V_0 = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t))$$

$$V_0(t) - L \frac{d(i_1(t) - i_2(t))}{dt} + R i_2(t) = 2 R i_1(t)$$

$$[i_1(t) = [V_0(t) = L \frac{d(i_1(t) - i_2(t))}{dt} + R i_2(t) \frac{1}{2R}]]$$

$$\underline{V_0(t) = R i_2(t) + \frac{1}{2} \int i_2(t) dt}$$